

Machine Learning

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Second Session - 17/02/2026 - **Duration:** 2 hours

Engineering in Computer Science and Artificial Intelligence

Candidates must answer in the designated spaces and submit *only* the official examination sheets; any additional materials will not be graded.

For the **multiple-choice questions**, exactly *one* of the four alternatives is correct. Correct answers receive one point, incorrect answers incur a penalty of 0.25, and unanswered questions have no effect on the score.

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Matricola: _____

1. Consider the task of optimizing the training error in logistic regression.
 - A. Stochastic Gradient Descent typically converges in fewer steps than Gradient Descent.
 - B. Implementing one step of gradient descent is faster than its stochastic counterpart.
 - C. A gradient estimation in stochastic gradient descent only uses one data point.
 - D. None of the above.
2. Logistic Regression is mainly used for
 - A. Minimizing the mean square error in linear regression.
 - B. Multi-class classification.
 - C. Solving SVM with the kernel trick.
 - D. Binary Classification.
3. The Gini Index is used to
 - A. Find the gradient step-size in Linear Regression.
 - B. Reduce overfitting in Linear Regression with Lasso Regularizer.
 - C. Balance the training of Decision Trees.
 - D. Implement the kernel Trick.
4. Consider the family of circles on the plane $\mathcal{H} = \{h_{a,b,r}(x_1, x_2) = \mathbb{I}_{(x_1-a)^2 + (x_2-b)^2 \leq r^2}, \text{ for } a, b, r \in \mathbb{R}\}$.
 - A. The VC dimension of $\mathcal{H}$ is 3.
 - B. The VC dimension of $\mathcal{H}$ is 4.
 - C. The VC dimension of $\mathcal{H}$ is 5.
 - D. The VC dimension of $\mathcal{H}$ is unbounded.
5. The primary application for Support Vector Machines is
 - A. Binary Classification.
 - B. Regression.
 - C. Clustering.
 - D. Reinforcement Learning.

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6. Adam is:
 - A. An algorithm for gradient computation more efficient than Backpropagation.
 - B. A neural network architecture.
 - C. An activation function that prevents the vanishing gradient problem.
 - D. An algorithm for training neural networks.
7. Gradient explosion may have the following consequences:
 - A. The model weights become very small, causing the learning process to stall or stop entirely.
 - B. The activation functions take tiny values close to zero, leading to precision loss.
 - C. The activation functions saturate, leading to overfitting.
 - D. None of the above.
8. What is the primary purpose of an Autoencoder's "bottleneck"?
 - A. To increase the dimensionality of the input data so that the model can find more complex features.
 - B. To force the network to learn a lower-dimensional representation of the input.
 - C. To provide a labeled output that the network can use for supervised classification tasks.
 - D. To store the entire input dataset in memory, ensuring perfect reconstruction when decoding.
9. A Softmax Layer:
 - A. Is an appropriate output layer in Neural Networks for multivariate regression.
 - B. Can be used to return a probability distribution.
 - C. Is used to limit the magnitude of activation functions' output.
 - D. None of the above.
10. k-Nearest Neighbors:
 - A. Is suitable for regression problems only for univariate functions.
 - B. Is very efficient at inference time.
 - C. Is suitable for classification problems.
 - D. None of the above.

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11. (5 points) Write the natural loss function for linear regression with Ridge Regularizer, and describe an algorithm for its minimization.

12. (6 points) Denote with $\sigma(\cdot)$ the sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$, and consider functions $h_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$ for $\theta \in \mathbb{R}^d$ defined as follows:

$$h_\theta(x) = \sigma(x^T \theta) = \frac{1}{1 + e^{-x^T \theta}}.$$

Write the log-likelihood $\hat{\ell}(\theta)$ for the generic h_θ on n points $(x^1, y^1), \dots, (x^n, y^n)$, where each (x^i, y^i) is in $\mathbb{R}^d \times \{0, 1\}$. Describe the gradient ascent and stochastic gradient ascent algorithms for the problem $\max_{\theta \in \mathbb{R}^d} \hat{\ell}(\theta)$. Why are the two gradient methods effective on this task?

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13. (5 points)

1. Illustrate the K-NN algorithm for Regression.
2. Provide an example of execution of 4-NN Regression for a function $f : \mathbb{R} \rightarrow \mathbb{R}$, reporting both the graphical and the analytical representation, and showing how the prediction on an unseen point $\bar{x}$ is obtained.

14. (6 points)

1. Define the architecture of an Autoencoder taking inputs from $\mathbb{R}^{16}$ and containing 2 hidden layers in both the encoder and the decoder, each halving/doubling the size of its input. Specify the type of the activation functions used in each layer. Provide **both**: (i.) the graphical and (ii.) the matrix representation of the architecture.
2. Report the loss function for the network defined above, assuming a training set containing N instances and considering the squared error.
3. Assume you are provided a network taking inputs from $\mathbb{R}^4$ but you need to process instances from $\mathbb{R}^{16}$. Illustrate how you could take advantage of the Autoencoder defined above.